

Jayasurya T Appu

Backend Engineer — Node.js — Microservices

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SUMMARY

Backend Engineer with 3+ years of experience building scalable, high-performance backend systems using Node.js, microservices, and event-driven architectures. Proven impact including reducing service cancellations by **80%**, improving system performance by up to **50%**, and supporting systems handling **~1000 requests/sec** in production. Strong focus on system design, reliability, and building production-grade distributed systems.

SKILLS

Backend & Systems: Node.js, Express.js, Python (Django, Flask), REST APIs, Microservices, gRPC, WebSockets

Databases & Caching: MongoDB, SQL, Redis (Caching & Performance Optimization)

Messaging & Streaming: Kafka (Event Streaming), RabbitMQ

DevOps & Infrastructure: Docker, Docker Compose, Nginx, GCP

Frontend (Supporting): React.js, Angular

Core Concepts: System Design, Distributed Systems, API Design, Performance Optimization, Scalability, High Availability

EXPERIENCE

Software Engineer — Carclenx (Pexa)

July 2025 – Present

Backend Engineer

- Architected and scaled backend systems for a real-time platform, handling **~1000 requests/sec** in production
- Built a **Worker & Staff Analytics engine**, reducing service cancellations by **~80%** through intelligent task allocation
- Designed a **distributed payroll system** for **200+ employees** using message queues, reducing manual workload by **70–80%**
- Replaced third-party WhatsApp automation with an **in-house event-driven system**, increasing lead generation by **~20%**
- Optimized backend workflows and database queries, improving API response times by **30–50%**
- Led a team of developers, owning architecture decisions and delivery

Junior Software Engineer — Quest Innovative Solution

Dec 2022 – Sept 2024

Backend Developer

- Developed **20+ production-grade APIs**, improving automation and reducing manual processes
- Built scalable backend services using **Node.js** and **Django** for business-critical applications
- Improved backend performance through optimized query design and efficient logic implementation
- Integrated frontend systems with backend APIs, ensuring reliable and efficient data flow

Full Stack Developer — Carclenx (Pexa)

Jan 2022 – June 2022

Software Developer

- Developed and maintained **RESTful APIs** and integrated third-party services for production applications
- Built and enhanced **admin dashboards using Angular**, enabling real-time data interaction with backend systems
- Improved application reliability by ensuring stable **frontend–backend communication** and efficient data flow

Backend Developer — Paco Studios

2025 – Present

Independent / Client Projects

- Delivered **production-grade backend systems**, handling end-to-end architecture and deployment
- Designed scalable APIs using **Node.js** and **MongoDB**
- Owned full project lifecycle: requirements, system design, development, and deployment

PROJECTS

Microservices E-commerce Backend System

- Designed a **distributed microservices architecture** with service isolation and **gRPC-based communication**
- Built an **event-driven system using Kafka**, enabling asynchronous processing and scalable service interaction
- Integrated **Redis caching** to reduce database load and improve API performance under high traffic
- Containerized services using **Docker** and orchestrated them with Docker Compose for consistent deployments
- Configured **Nginx as a reverse proxy/API gateway** for routing, load balancing, and request handling
- Designed the system for **scalability, fault tolerance, and production readiness**
- **Tech Stack:** Node.js, Express, Kafka, Redis, gRPC, Docker, Nginx

AI Chat System with RAG (Retrieval-Augmented Generation)

- Built a **RAG-based backend system** using FastAPI, ChromaDB, and SentenceTransformers for semantic search
- Implemented **vector embeddings and retrieval pipelines** to enable context-aware responses
- Integrated a local **LLM (Mistral)** for generating intelligent answers based on retrieved data
- Designed a pipeline for **document ingestion (PDF → embeddings → query → response)**
- Architected the system for **scalable AI-assisted querying** and knowledge retrieval

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science
Core Computer Science Curriculum Completed

2017 – 2021